

# Data Cleaning and Exploratory Data Analysis Summary Report

## Project Overview

This project involved cleaning and analyzing the Online Retail dataset using Python and Pandas. The objective was to improve data quality, explore customer purchasing patterns, and generate meaningful business insights through exploratory data analysis and data visualization.

## Cleaning Challenges Encountered

Several data quality issues were identified during the cleaning process.

- Missing values were found in the **Description** and **CustomerID** columns.
- Duplicate records were detected and removed.
- Some transactions contained negative quantities and zero unit prices, which were considered invalid and removed.
- The **InvoiceDate** column was converted from text format to datetime format.
- Column names were standardized by converting them to lowercase for consistency.

These steps improved the accuracy and reliability of the dataset before analysis.

## Key Exploratory Data Analysis Findings

The exploratory analysis revealed several important patterns:

- **PAPER CRAFT, LITTLE BIRDIE** was the highest-selling product.
- The **United Kingdom** generated the highest total revenue among all countries.
- Sales increased throughout 2011 and reached their highest level in **November 2011**.
- A small group of customers purchased significantly larger quantities than others.
- Most products were sold at relatively low prices, while only a few products had high unit prices.

## Top Insights

1. The United Kingdom is the company's largest market and contributes the majority of total revenue.
2. A few products account for a substantial proportion of total sales volume.
3. Sales show clear seasonal trends, with stronger performance during the final months of the year.

4. A small number of customers make very large purchases and represent valuable customers.
5. Most products are low-priced, indicating that the business focuses primarily on affordable retail items.

## **Conclusion**

The data cleaning process successfully improved the quality of the dataset by addressing missing values, duplicate records, invalid entries, and formatting inconsistencies. The exploratory data analysis provided valuable insights into customer behavior, product performance, revenue distribution, and sales trends. These findings can support better business decisions in areas such as inventory management, marketing, and customer relationship management.